REGRESSION EQUATIONS

Regression analysis is a mathematical measure of the average relationship between two or more variables.
In regression analysis there are two types of variables.
The variable whose value is to be predicted is called dependent variable and the variable whose value is used for prediction, is called the independent variable.

In case of two variables x and y we shall have two lines of regression; one of y on x and the other of x on y

The line of regression of y on x is the line which gives the best estimate for the value of y for any specified value of x.

Similarly, the line on regression of x on y is the line which gives the best estimate for the value x for any specified value of y.

The techniques of deriving the two regression equations is explain below:

Equation of regression line of y on x

$$y = a + bx$$

$$b = \frac{\sum dxdy}{\sum dx^2} \qquad a = \bar{y} - b\bar{x}$$

Alternatively, $y - \bar{y} = b\left(x - \bar{x}\right)$

b is the gradient of the line and is known as the regression coefficient of y on x
a is the y-intercept

The equation of regression line of x on y

$$x = c + dy$$

Where, $c = \bar{x} - d\bar{y}$, $d = \dfrac{\sum dxdy}{\sum dy^2}$

Also, since the line passes through $\left(\bar{x}, \bar{y}\right)$

$$x - \bar{x} = d\left(y - \bar{y}\right)$$

d is known as the regression coefficient of x on y.

$r^2 = b \times d$ implying, $r = \pm\sqrt{b \times d}$

$\ell = 1 - \dfrac{6\sum d^2}{n(n^2 - 1)}$, where d=rank of x less rank of y

And n= no. of pairs

EXAMPLE

From the data given below find:

a) The two regression equations
b) The coefficient of correlation between the marks in Economics and Statistics
c) The most likely marks in Statistics when marks in Economics are 30

Marks in Economics (x)	25	28	35	32	31	36	29	38	34	32
Marks in Statistics (y)	43	46	49	41	36	32	31	30	33	39

SOLUTION

Let us denote the marks in Economics by the variable X and the marks in Statistics by the variable Y

CALCULATIONS FOR REGRESSION EQUATIONS

X	Y	$dx=x-\bar{x}$ $=x-32$	$dy=y-\bar{y}$ $=y-38$	dx^2	dy^2	dxdy
25	43	-7	5	49	25	-35
28	46	-4	8	16	64	-32
35	49	3	11	9	121	33
32	41	0	3	0	9	0
31	36	-1	-2	1	4	2
36	32	4	-6	16	36	-24

29	31	-3	-7	9	49	21
38	30	6	-8	36	64	-48
34	33	2	-5	4	25	-10
32	39	0	1	0	1	0
$\sum x=320$	$\sum y=380$	$\sum dx=0$	$\sum dy=0$	$\sum dx^2=140$	$\sum dy^2=398$	$\sum dxdy=-93$

Here,

$$\bar{x} = \frac{\sum x}{n} = \frac{320}{10} = 32;\ \bar{y} = \frac{\sum y}{n} = \frac{380}{10} = 38$$

Coefficient of regression of y on x is given by

$$b_{yx}=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \frac{\sum dxdy}{\sum dx^2} = \frac{-93}{140} = -0.6643$$

Similarly, coefficient on regression of x on y is

$$b_{xy}=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(y-\bar{y})^2} = \frac{\sum dxdy}{\sum dy^2} = \frac{-93}{398} = -0.2337$$

Equation of the line of regression of y on x is:

$$y - \bar{y} = b_{yx}(x - \bar{x})$$
$$y - 38 = -0.6643(x - 32)$$
$$y = -0.6643x + 38 + 0.6643 \times 32$$
$$y = -0.6643x + 59.2576$$

Equation of the line of regression of x on y is:

$$x - \bar{x} = b_{xy}(y - \bar{y})$$
$$x - 32 = -0.2337(y - 38)$$
$$x = -0.2337y + 0.2337 \times 38$$
$$x = -0.2337y + 40.8806$$

Correlation coefficient

We have

$$r^2 = b_{yx}b_{xy} = (-0.6643) \times (-0.2337) = 0.1552$$
$$r = \pm\sqrt{0.1552} = \pm 0.394$$

Since both the regression coefficients are negative, r must be negative. Thus, discarding the plus sign, we have

r=-0.394

In order to estimate the most likely marks in Statistics (y) when marks in Economics (x) are 30, we shall use line of regression of y on x:

$$y = -0.6643 \times 30 + 59.2576 = 39.3286$$

Hence, the most likely marks in Statistics when marks in Economics are 30, are

$$\simeq 39$$

HOMEWORK

Obtain the equations of the two lines of regressions for the following data:

X	43	44	46	40	44	42	45	42	38	40	42	57
y	29	31	19	18	19	27	27	29	41	30	26	10

Hence obtain the value of the correlation coefficient between X and Y

ASSGT

The yield of a particular crop on a farm is thought to depend principally on the amount of rainfall in the growing season. The values of the yield y, in tons per acre and the rainfall x. in centimeters, for seven successive years are given in the following table:

X	12	14	15	11	13	14	12
Y	6	8	8	5	7	9	6

(a) Find the equation of the least squares regression line of

(i) y on x

(ii) x on y

(b) Given that the rainfall in the growing season of a subsequent year was 14 cm, estimate the yield in that year

c) Given that the yield in a subsequent year was 8 tons per acre, estimate the rainfall in the growing season of that year